Tae Yong Song

Ph.D. Student in Building Science
Building Simulation Lab (BS Lab)
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EDUCATION

2026.03–present **Ph.D. in Architecture and Architectural Engineering**, Seoul National University
Advisor: Prof. Cheol-Soo Park

2022.03–2024.02 **M.S. in Architecture and Architectural Engineering**, Seoul National University
Advisor: Prof. Cheol-Soo Park
Thesis: "Exploring Potential of Transfer Learning in Overcoming Data Scarcity for Building Energy Model"

2015.03–2022.02 **B.S. in Architecture and Architectural Engineering**, Seoul National University

PUBLICATIONS

International Journal Article

2025 Physics-embedded hybrid modelling approach for room temperature prediction using Siamese neural network and RC model

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, Junu Lee, and Cheol-Soo Park. *Journal of Building Performance Simulation*, 19(2), 277–288. Published online 5 August 2025; issue dated 2026.

International Conference Papers

2026 Curated Intelligence for Visual Inference and Deterministic Logic toward Automatic BEM

Tae Yong Song, Jin-Hong Kim, Seung-Ju Lee, Soo-Kyung Kim, Hyun-Chul Park, Kyung-Jae Kim, and Cheol-Soo Park. *Proceedings of the 7th Building Simulation Applications Conference*, June 24–26, Bozen-Bolzano, Italy.

2024 Physics-Informed Hybrid Modeling Approach for Room Temperature Prediction Using an RC Model and Siamese Neural Network

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, and Cheol-Soo Park. *SimBuild 2024*.

2023 Causality-based model for decision-making of building energy retrofit

Chul-Hong Park, Young-Sub Kim, **Tae Yong Song**, Seon-Jung Ra, and Cheol-Soo Park. *Building Simulation 2023*, 1433–1438.

2023 Limitations of Conventional Artificial Neural Network-Based Surrogate Models for Building Energy Retrofit

Chul-Hong Park, Young-Sub Kim, Seon-Jung Ra, **Tae Yong Song**, and Cheol-Soo Park. *ASHRAE Annual Conference*, 683–690.

Domestic Conference Papers

2026 Development of Customized AI Technologies for BIM-to-BEM Automation: Integration of LLMs, Computer Vision, Delaunay Geometry Algorithms, and Minimum Cycle Algorithms

Tae Yong Song, Jin-Hong Kim, Soo-Kyung Kim, Hyun-Chul Park, Kyung-Jae Kim, and Cheol-Soo Park. *Proceedings of the 2026 Spring Annual Conference of the Architectural Institute of Korea*, 46(1), 829–830. **Outstanding Presentation Award.**

2022 Dynamic Sensitivity Analysis Using NODE-GAM

Tae Yong Song, Chul-Hong Park, Seon-Jung Ra, Young-Sub Kim, and Cheol-Soo Park. *Proceedings of the 2022 Annual Conference of the Korean Institute of Architectural Sustainable Environment and Building Systems*, 181–182.

RESEARCH PROJECTS

2026.03–2030.12 **Virtual Building Modeling and Simulation Framework, and a Causality-Based AR Tool for Net-Zero Building Design Evaluation**
Korea Institute of Civil Engineering and Building Technology (KICT)

2025.10–2026.04 **LLM Technology for Digital-Twin-Based Building Energy Models**
Samsung Electronics

2025.12–2026.01 **IDF Model Development for Digital-Twin Energy Simulation**
Eterion (Samsung Electronics)

2022.04–2024.12 **Automated Digital Diagnosis and Design Technologies for Green Remodeling**
Korea Agency for Infrastructure Technology Advancement (KAIA)

2023.05–2023.11 **Data-Driven HVAC Load Prediction and Optimal Air-Handling-Unit Control**
LG Electronics

2022.12–2023.07 **Greenhouse-Gas Reduction Factors and Emissions Assessment for Building Green Remodeling**
Ministry of Land, Infrastructure and Transport (MOLIT)

TEACHING EXPERIENCE

2026–present **Teaching Assistant**, Seoul National University
Department of Architecture and Architectural Engineering

2023 **Teaching Assistant**, Seoul National University
Department of Architecture and Architectural Engineering

HONOR

2026 **Outstanding Presentation Award**
Architectural Institute of Korea Spring Annual Conference, Oral Presentation – General Division

Last updated: July 2026 Full publication details and project updates: taeyong-song.github.io